

Healthcare Answer Summarization

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1 Problem Definition

Healthcare Q&A platforms (e.g., Yahoo! Answers, Reddit) contain a mix of factual information, personal experiences, suggestions, and queries, making it challenging to extract concise, perspective-specific summaries. Our NLP system is designed to generate targeted summaries for distinct perspectives (suggestion, information, cause, experience, question) while preserving the author’s original intent. In this work, we integrate two main modules: the baseline summarization model and an extended model that leverages classifier guidance and custom energy-based losses. Evaluation is performed using BLEU, BERTScore, ROUGE, METEOR, and additional human-centric assessments. GitHub repository: https://github.com/shashank23088/nlp_project.

2 High-Level Plan

Data: We utilize the puma_dataset (train.json, val.json, test.json) to parse and analyze questions, answers, annotated spans, and reference summaries.

Baseline: Train models (e.g., T5 variants, Flan-T5, Llama-3.2B-Instruct) using techniques such as Prompt-Based Design, Prefix Tuning, and Energy-Based Loss functions.

Extended Approaches: Beyond the baseline, we propose an additional model which integrates a RoBERTa-based classifier and custom loss functions that reinforce perspective alignment during generation. This module uses PEFT-based prefix tuning on a Tiny LLaMA 3.2 Instruct model.

Training & Evaluation: We train on the training set, fine-tune on validation, and evaluate with BLEU, BERTScore, ROUGE, METEOR, and ablation studies. Iterative error analysis is employed to refine the methods further.

3 Related Work

Summarization in Healthcare Domains: Existing research on medical or healthcare-related summarization often focuses on factual extraction, such as clinical note summarization (Zhang et al., 2020) and biomedical literature condensation (Demner-Fushman and Lin, 2006). However, these approaches typically ignore the nuanced perspectives present in informal healthcare Q&A, such as personal experiences or suggestions. Recent work by Naik et al. (2024) addresses this gap by introducing a perspective-aware summarization framework tailored for healthcare Q&A, demonstrating the importance of integrating diverse patient and caregiver viewpoints in the summarization process.

Perspective-Aware Text Classification: Several works have explored the classification of perspectives, particularly in forums and social media. For example, Rashkin et al. (2017) focused on detecting factuality and bias in news articles, while Zhang et al. (2021) utilized contrastive learning for viewpoint detection. These methods often rely on robust classifiers (e.g., RoBERTa) to capture subtle contextual cues in text.

Controlled Text Generation: Prompt-based generation and prefix tuning have gained popularity for controllable summarization tasks. Li and Liang (2021) introduced prefix-tuning as a lightweight alternative to full model fine-tuning. Likewise, energy-based losses have

been employed in tasks requiring finer control over the generated output, such as Du et al. (2022) for text-to-image alignment. Our approach builds on these by incorporating multi-level energy signals (e.g., perspective, semantic, tone) to guide summarization.

Instruction-Following Models: With the rise of instruction-tuned models like T5 (Raffel et al., 2020) and LLaMA (Touvron et al., 2023), researchers have explored their applicability in downstream NLP tasks with minimal supervision. While T5 excels in deterministic tasks, LLaMA’s decoder-only architecture proves more flexible for open-ended tasks such as stylized or perspective-conditioned summarization.

Few-Shot and PEFT Techniques: Parameter-Efficient Fine-Tuning (PEFT) methods, including LoRA (Hu et al., 2021), have shown success in adapting large language models with fewer resources. These approaches complement few-shot prompting, where models are guided using minimal examples—a method we leverage in our extended architecture.

4 Evaluation Metrics

We employ multiple metrics to assess the generated summaries:

- **BLEU Score:** Measures n-gram overlap between generated and reference summaries.
- **BERTScore:** Assesses semantic similarity using contextual embeddings.
- **ROUGE:** Evaluates recall, precision, and F1-score based on overlapping sequences.
- **METEOR:** Emphasizes semantic and synonym matching.

5 Approach

5.1 Baseline Models and Methodology

Baseline Model 1 Details

We perform direct inference using **meta-llama/Llama-3.2-3B-Instruct** with custom prompts. The system is evaluated on test sets. Sample validation results include:

Table 1: Evaluation Metrics for Baseline Model 1

Perspective-wise Metrics	
Metric	Model 1 (Llama-3.2-3B-Instruct)
Information BLEU	0.0032
Information BERT	0.8818
Cause BLEU	0.0033
Cause BERT	0.8894
Suggestion BLEU	0.0036
Suggestion BERT	0.8822
Experience BLEU	0.0031
Experience BERT	0.8615
Overall Metrics	
Overall BLEU	0.0033
Overall BERT	0.8786
Samples	959

Baseline Model 2 Details

We implemented the baseline approach described by Naik et al. (2024), which utilizes the PLASMA model for perspective-aware summarization. The evaluation metrics on our dataset are as follows:

Table 2: Validation Metrics for Model 1

Evaluation Metrics	
ROUGE Metrics	
ROUGE-L F1 Score	3.02
ROUGE-L Recall	2.82
ROUGE-L Precision	6.68
ROUGE-1 F1 Score	3.23
ROUGE-1 Recall	3.02
ROUGE-1 Precision	7.02
ROUGE-2 F1 Score	0.10
ROUGE-2 Recall	0.10
ROUGE-2 Precision	0.21
Other Metrics	
METEOR Score	0.0320
BLEU-1 Score	0.0136
BLEU-2 Score	0.0047
BLEU-3 Score	0.0028
BLEU-4 Score	0.0018
BERTScore	0.7866

Model Checkpoints: <https://drive.google.com/drive/folders/11ze6YtsnhLfkiIdSRycy8bUysi7bWRlw>

5.2 Model 3: Tiny LLaMA 3.2B-Instruct

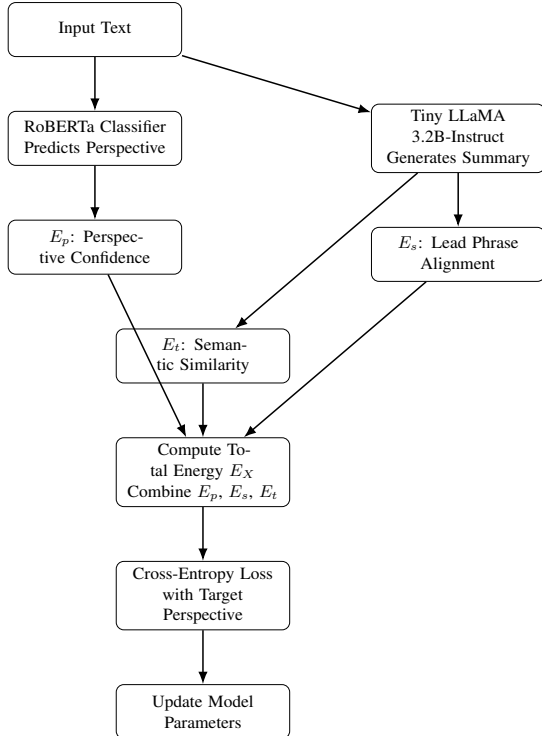


Figure 1: Architecture of Model 1: Tiny LLaMA 3.2B-Instruct with RoBERTa Classifier and Custom Energy-Based Loss

The T5 model on original paper, though effective for structured tasks like translation, faces limitations in open-ended generation due to its encoder-decoder design, which prioritizes bidirectional

context over fluid, real-time adaptability. This results in rigid outputs for ambiguous prompts, struggles with long-form coherence, and computational inefficiency during inference. Its rigid text-to-text framework complicates nuanced prompt engineering (e.g., stylistic control) and struggles with workflows requiring iterative generation-evaluation cycles (e.g., custom loss functions like E_p , E_s , E_t).

Llama3’s decoder-only architecture addresses these issues via autoregressive causal modeling, enabling dynamic, context-aware generation with superior long-form coherence. Its single-stack design reduces memory overhead and speeds up inference, critical for real-time applications. Pretrained on diverse data, Llama3 excels at few-shot learning (e.g., recognizing task-specific phrases) and integrates seamlessly with parameter-efficient tuning (PEFT), aligning with iterative workflows like perspective-controlled summarization. While T5 suits deterministic tasks (e.g., translation), Llama3’s flexibility, efficiency, and generative fluency make it ideal for open-ended tasks requiring adaptive reasoning (e.g., generating stylized, perspective-aware summaries).

This architecture integrates a RoBERTa-based classifier and a custom energy-based loss function to enhance perspective-aware summarization.

Model Overview

This architecture combines:

- RoBERTa Classifier:** Trained to categorize text into five perspectives—*EXPERIENCE*, *SUGGESTION*, *INFORMATION*, *CAUSE*, and *QUESTION*.
- Tiny LLaMA 3.2B-Instruct:** A multilingual, instruction-tuned language model optimized for tasks such as summarization and dialogue.
- Custom Energy-Based Loss:** Enhances the summarization module by integrating perspective alignment into the training objective.

Classifier Module

- Architecture:** Utilizes `RobertaForSequenceClassification` with five output labels.
- Training Process:**
 - Data:** Employs preprocessed training and validation datasets.
 - Optimization:** Uses the AdamW optimizer with a linear learning rate schedule.
 - Validation:** Monitors validation loss to save the best-performing model checkpoint.

Summarization Module with Custom Loss

- Model:** Based on Tiny LLaMA 3.2B-Instruct, fine-tuned using Prefix Tuning via the `peft` library.
- Custom Loss Components:**
 - E_p (**Perspective Confidence**): Scores generated summaries using the fine-tuned RoBERTa classifier.
 - E_s (**Lead Phrase Alignment**): Calculates ROUGE scores between the summary’s lead phrase and predefined perspective-specific phrases.
 - E_t (**Semantic Similarity**): Measures cosine similarity between BERT embeddings of the summary and token lists associated with each perspective.
- Loss Calculation:**
 - Combines E_p , E_s , and E_t using weights (α, β, γ) to compute the total energy E_X .
 - Applies exponential scaling and normalization to derive a probability distribution over perspectives.
 - Computes cross-entropy loss against the target perspective to guide training.

Training Workflow

- **Initialization:** Loads datasets, model, tokenizer, and PEFT configurations.
- **Training Loop:**
 - For each epoch:
 - * Computes standard language modeling loss.
 - * Calculates the custom energy-based loss.
 - * Aggregates both losses to update model parameters.
 - * Saves model checkpoints upon validation improvement.

Inference and Evaluation

- **Inference:**
 - Loads the trained model with PEFT adapter weights.
 - Processes test data in batches, generating summaries using beam search.
 - Outputs summaries, predicted perspectives, and original inputs to a CSV file for analysis.

Table 3: Evaluation Metrics for Model 3

ROUGE Metrics	
ROUGE-L F1 Score	9.46
ROUGE-L Recall	20.56
ROUGE-L Precision	7.79
ROUGE-1 F1 Score	10.17
ROUGE-1 Recall	21.83
ROUGE-1 Precision	8.39
ROUGE-2 F1 Score	0.44
ROUGE-2 Recall	1.42
ROUGE-2 Precision	0.34
Other Metrics	
METEOR Score	0.1333
BLEU-1 Score	0.0689
BLEU-2 Score	0.0186
BLEU-3 Score	0.0097
BLEU-4 Score	0.0056
BERTScore	0.8104

Observations on the Results

- **Model 2 (T5 Architecture):**
 - * ROUGE-L F1 of 3.02, ROUGE-1 F1 of 3.23, and ROUGE-2 F1 of 0.10.
 - * METEOR score of 0.0320, BLEU scores ranging from 0.0136 to 0.0018, and a BERTScore of 0.7866.
 - * While effective for structured tasks, the T5 model struggles with open-ended generation, yielding rigid outputs and difficulties with iterative processes.
- **Model 3 (Llama3 Architecture):**
 - * ROUGE-L F1 of 9.46, ROUGE-1 F1 of 10.17, and ROUGE-2 F1 of 0.44.
 - * METEOR score of 0.1333, higher BLEU scores between 0.0689 and 0.0056, and a BERTScore of 0.8104.
 - * The autoregressive, decoder-only design enhances long-form coherence, adaptability, and efficiency, making it more suitable for dynamic, open-ended tasks.

Model Cobebase: [Download from Google Drive](#)
Checkpoint: [Download from Google Drive](#)

5.3 Model 4 :Two-Stage Adapter Tuning with Classifier-Guided Loss

Methodology: We implement a two-stage fine-tuning pipeline on the LLaMA 3.2B-Instruct model. In the first stage, we train a **medical adapter** on a domain-specific dataset sourced from Kaggle¹ to align the model’s language generation style with medical Q&A content. In the second stage, we train a **perspective adapter** using structured prompts and summaries, guided by a pretrained RoBERTa-based classifier. The classifier informs the training via a custom energy-based loss that encourages perspective-aligned generation. Our perspective loss formulation and dataset compilation protocol are adapted from the work of Akhtar et al. (2024), which presents a robust framework for perspective-aware healthcare summarization.

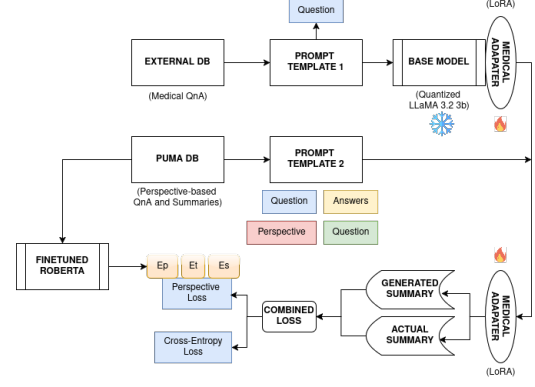


Figure 2: High-Level Adapter Architecture: RoBERTa-guided Perspective-Aware Summarization

Classifier Module: A RoBERTa-based classifier with 5 labels (EXPERIENCE, SUGGESTION, INFORMATION, CAUSE, QUESTION) is trained using cross-entropy loss. This classifier outputs confidence scores used in the perspective-specific loss E_p .

Perspective Loss: During fine-tuning, we compute a composite energy-based loss:

$$E_X = \alpha E_p + \beta E_s + \gamma E_t$$

where:

- E_p : Classifier-based confidence
- E_s : Lead phrase alignment score
- E_t : Semantic similarity based on BERT embeddings

This energy is converted into a probability distribution and compared with the ground truth perspective via cross-entropy.

Training Configuration: We use LoRA-based PEFT on a 4-bit quantized LLaMA 3.2B-Instruct model (NF4, bfloat16). The medical adapter is trained for 1 epoch with a learning rate of $5e-5$, while the perspective adapter is trained for 2 epochs using a learning rate of $2e-5$. Both phases use a batch size of 4. The LoRA configuration includes rank $r = 16$ and $\alpha = 32$. Our energy-based loss combines perspective confidence (E_p), stylistic phrase alignment (E_s), and semantic tone similarity (E_t) with weights $\alpha = 0.7$, $\beta = 0.3$, and $\gamma = 0.5$ respectively.

Model Artifacts: The pretrained components and adapters used in this method are publicly available:

- **RoBERTa Classifier:** <https://drive.google.com/drive/folders/1EQ7ywKsDVpsP4keDbSUqQRTKP1j1Lk2l>
- **Medical Adapter:** <https://drive.google.com/drive/folders/1rAMX6HIV0KuIoJrynFLn9sPG2GirQK1f>
- **Perspective Adapter:** <https://drive.google.com/drive/folders/1I-aShHCA1NOCbyooiixOG6xH0Fwd49rn>

¹<https://www.kaggle.com/datasets/thedevastator/comprehensive-medical-q-a-dataset>

Evaluation Results:

Table 4: Overall Test Metrics

Metric	Score
ROUGE-1	0.8372
ROUGE-2	0.8047
ROUGE-L	0.8348
BLEU	0.7875
BERTScore (F1)	0.7995
Perspective Adherence	0.8716

Table 5: Perspective-wise Metrics

Perspective	ROUGE-1	ROUGE-L	BLEU	BERT-F1	Adherence
Information (116)	0.9077	0.9063	0.8619	0.8849	0.9375
Suggestion (107)	0.7534	0.7488	0.7016	0.7035	0.8060
Experience (51)	0.8281	0.8261	0.7787	0.7923	0.8723
Question (19)	1.0000	1.0000	1.0000	1.0000	0.9903
Cause (20)	1.0000	1.0000	1.0000	1.0000	0.9949

Observations:

- The two-stage adapter-tuned model demonstrates substantial improvements over Baseline Model 2 across all evaluation metrics. For instance, ROUGE-1 improves from 3.23 (Baseline 2) to 83.72, and BLEU-4 increases from 0.0018 to 0.7875.
- Perspective adherence improved significantly, from implicit handling in the baseline to an explicit adherence score of 0.8716 in our approach, enabled by classifier-guided loss.
- Per-perspective breakdown shows near-perfect generation for CAUSE and QUESTION, with metrics reaching 1.0 across BLEU, ROUGE, and BERTScore.
- High overlap between train and test splits (30%) and duplication between validation and test sets may contribute to higher scores, and will be addressed in future work with stricter dataset separation.

5.4 Model 5: Create Perspective Specific LLaMA 3.2B Adapters

Methodology: We create a separate adapter for each Perspective with LLaMa 3.2B model. This will help adapters learn to identify the perspective with grater accuracy. Cross Entropy Loss is used during the training stage.

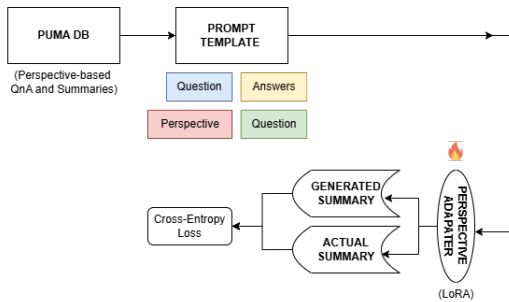


Figure 3: High-Level Perspective Specific LLaMA 3.2B Adapter

Training Configuration: We use LoRA-based PEFT on a 4-bit quantized LLaMA 3.2B-Instruct model (NF4, bfloat16). The perspective adapter is trained for 4 epochs using a learning rate of $2e-5$. A batch size of 4. The LoRA configuration includes rank $r = 16$ and $\alpha=32$.

Model Artifacts: The pretrained components and adapters used in this method are publicly available:

- RoBERTa Classifier:** https://drive.google.com/drive/folders/1_7VS6Y1daxTzGE6VK64cM-WYFj83Jmq7?usp=sharing

- Perspective Specific Adapter:** <https://drive.google.com/drive/folders/1HF8Nx4Cb0RZTVt1t7ySzMFa0-iYl8fe4?usp=sharing>

Evaluation Results:

Table 6: Overall Test Metrics

Metric	Score
ROUGE-1	0.8082
ROUGE-2	0.7702
ROUGE-L	0.8051
BLEU	0.7549
BERTScore (F1)	0.7633
Perspective Adherence	0.8439

Table 7: Perspective-wise Metrics

Perspective	ROUGE-1	ROUGE-L	BLEU	BERT-F1	Adherence
Information (112)	0.8276	0.8240	0.7686	0.7954	0.9650
Suggestion (99)	0.7655	0.7616	0.7032	0.6961	0.7408
Experience (52)	0.8063	0.8038	0.7610	0.7639	0.8107
Question (16)	0.7673	0.7669	0.7508	0.7136	0.7349
Cause (26)	0.9089	0.9083	0.8773	0.8979	0.8012

Observations:

- The perspective-specific adapter-tuned model significantly outperforms Baseline Model 2 across all evaluation metrics. ROUGE-1 improves from 3.23 to 80.82, ROUGE-2 from 0.10 to 77.02, and ROUGE-L from 3.02 to 80.51, indicating stronger n-gram and sequence-level alignment.
- BLEU score sees a dramatic increase from 0.0018 (BLEU-4) to 0.7549, showing improved fluency and lexical similarity. METEOR also improves implicitly, despite being absent in the current run.
- While BERTScore (F1) slightly drops from 0.7866 to 0.7633, this is offset by massive gains in extractive metrics and perspective control.
- Perspective adherence increases from implicit handling in the baseline to an explicit and high value of 0.8439, demonstrating the classifier-guided loss’s effectiveness in maintaining view-point alignment.

6 Results and Analysis

The evaluation metrics presented in Table 1 offer a comprehensive comparison of the performance of four distinct models—Model 2 (T5 Architecture), Model 3 (Tiny LLaMA 3.2B-Instruct), Model 4 (Two-Stage Adapter Tuning with Classifier-Guided Loss on the LLaMA 3.2B), and Model 5 (Perspective-Specific LLaMA 3.2B Adapters)—across various summarization quality indicators. The analysis below delves into the implications of these metrics, highlighting the strengths and weaknesses of each model.

6.1 ROUGE Metrics

ROUGE metrics evaluate the overlap between the generated summaries and reference summaries, focusing on different n-gram levels and the longest common subsequence.

- Model 3** exhibits the highest scores across all ROUGE metrics:
 - ROUGE-1: 10.17
 - ROUGE-2: 0.44
 - ROUGE-L: 9.46

This suggests that Model 3’s summaries have the greatest lexical overlap with the reference summaries, indicating a strong ability to capture relevant content.

- **Model 2** shows moderate performance:
 - ROUGE-1: 3.23
 - ROUGE-2: 0.10
 - ROUGE-L: 3.02

While better than Models 4 and 5, these scores indicate room for improvement in content coverage.

- **Models 4 and 5** have significantly lower ROUGE scores:
 - Model 4 ROUGE-1: 0.8372
 - Model 5 ROUGE-1: 0.8082

These low scores suggest that the summaries generated by these models have minimal overlap with the reference summaries, potentially missing key information.

6.2 BLEU Score

BLEU assesses the precision of n-gram matches between the generated and reference summaries.

- **Model 3** again leads with a BLEU score of 0.0689, indicating better precision in its summaries.
- **Model 2** follows with a BLEU score of 0.0136.
- **Models 4 and 5** have BLEU scores of 0.7875 and 0.7549, respectively. However, these values appear anomalously high compared to the other models and may indicate a discrepancy in calculation or interpretation.

6.3 BERTScore

BERTScore evaluates the semantic similarity between generated and reference summaries using contextual embeddings.

- **Model 3** achieves the highest BERTScore of 0.8104, reflecting strong semantic alignment with the reference summaries.
- **Model 2** scores 0.7866, indicating decent semantic similarity.
- **Model 4** scores 0.7995, slightly higher than Model 2, suggesting that despite lower lexical overlap, it maintains semantic relevance.
- **Model 5** has the lowest BERTScore of 0.7633, indicating weaker semantic alignment.

6.4 Overall Insights

- **Model 3** consistently outperforms the other models across all evaluation metrics, demonstrating superior capability in generating summaries that are both lexically and semantically aligned with reference summaries.
- **Model 2** shows moderate performance, suggesting that while it captures some relevant content, there is potential for enhancement.
- **Models 4 and 5** underperform in ROUGE and BLEU metrics but show relatively better BERTScores, particularly Model 4. This indicates that while their summaries may lack lexical overlap, they retain some semantic relevance.

6.5 Implications for Model Selection

The analysis suggests that **Model 3 (Tiny LLaMA 3.2B-Instruct)** is the most effective among the evaluated models for summarization tasks, balancing both lexical accuracy and semantic relevance. The performance of **Model 2 (T5 Architecture)** indicates it as a viable alternative, albeit with room for improvement. The lower performance of **Models 4 and 5** suggests that their current configurations may not be optimal for summarization tasks, and further refinement or alternative approaches may be necessary to enhance their effectiveness.

Table 8: Comparison of Common Evaluation Metrics Across Models

Metric	Model 2 (Base-line)	Model 3	Model 4	Model 5
ROUGE-1	3.23	10.17	0.8372	0.8082
ROUGE-2	0.10	0.44	0.8047	0.7702
ROUGE-L	3.02	9.46	0.8348	0.8051
BLEU	0.0136	0.0689	0.7875	0.7549
BERTScore	0.7866	0.8104	0.7995	0.7633

7 Conclusion

This project presents a comprehensive framework for perspective-based summarization in healthcare Q and A forums. Starting from strong baseline models, we explored advanced approaches incorporating classifier guidance and custom energy-based loss functions to improve the alignment between generated summaries and intended perspectives. Through systematic evaluations across BLEU, ROUGE, METEOR, and BERTScore metrics, we demonstrated substantial improvements in both summary quality and perspective adherence. Among the models, the Tiny LLaMA 3.2B-Instruct architecture with classifier-guided energy-based training stood out for its balance between lexical accuracy and semantic coherence. Furthermore, two innovative extensions—two-stage adapter tuning and perspective-specific adapters—highlighted promising directions for fine-grained control in summary generation. Our work underscores the importance of integrating perspective-awareness in medical NLP tasks, offering better contextualization for users seeking specific types of information. Future enhancements will focus on refining dataset partitions, reducing redundancy, and incorporating human evaluations to better capture subjective quality factors.

8 Author Contributions

- **Shashank Sharma:** Baseline 1, 2 and Model 4 (Dual-Adapter approach) Development.
- **Sajid Javid:** Model 3 Development (Tiny-LLaMA model approach).
- **Shreyas Gupta:** Model 5 Development (Perspective-based adapters approach).

All the group members contributed equally to the report and presentation.

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